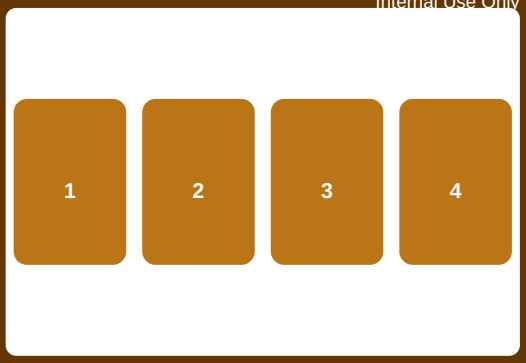
OPERATIONS

Early Product Development (EPD)

Apply lean and concurrent engineering principles to the development phase to shorten cycles, reduce engineering



WHEN TO USE

When product launches are consistently late, over budget, or followed by high defect rates in early production.

HOW TO APPLY — STEPS

1. Define customer value explicitly before designing any technical solution
2. Use set-based concurrent engineering — develop multiple options in parallel
3. Apply obeya (visual project rooms) to make progress and decisions visible
4. Conduct regular technical reviews with the chief engineer or equivalent
5. Apply FMEA and Design for X at the design stage — not after tooling is committed

PRACTITIONER TIP

! The cost of changing a design doubles at every gate. Fix problems on paper and simulation, not in tooling.

KEY OUTPUTS

- Obeya room with live project status
- Set-based design options comparison
- Engineering FMEA
- Knowledge-based decisions log

HOW TO MEASURE SUCCESS

Engineering change orders post-launch; time-to-market vs plan; first-time quality in production launch month

TOOL CATEGORY
Operations